

Experiments : Study of LAN Topology, Network Device and Address Configuration.

Problem Statement

1. Draw the Network Topology:

- Draw and identify LAN topology of your organisation and label all components such as Router, Switch, Access Point and end devices.

→ Below is detail case study of campus network.

- Layer 3 core switches in server room.
- Managed by POE + switch.
- Wireless Access point
- Multiple Ethernet port in class & lab.

* Network Topology

Hybrid Hierarchical topology which combines

- Tree topology
- Extended star topology
- Ring topology (server room)

→ Network Hierarchical layer

i) core layer - It contains server room and server firewall, Router, 2-L3+ switches, server, inverter etc.

ii) Distribution layer - Here, it contains 4 switches.

- (i) Main Build → R7 → (2nd floor)
- (ii) Main Build → R1 → (3rd floor)
- (iii) Main Build → R6 → (1st floor)
- (iv) Main Build → R2 → (2nd floor)

And, each contains 4 switches:-

① 18-port Gigabit easy smart switch with 16-port. TL-SG1218MPE

② SG3428XMP - 24-port Gigabit & 4-port 10 GbE SFP +, L2+ Managed switches with 24 ports.

* SG1218MPE → 18-port gigabit "Easy smart" Switch. Supports 16 port.

→ Used for CCTV, wifi etc.

* SG3428XMP → 24 POE + port uplinks, 10G SFP, usually act as main distribution point for each floor.

iii> Access Layer → It consists of 3-switches.

① R1 [Room - 301, 3rd floor]

② R4 [Room - 05, 1st floor]

③ R5 [Room - 18, 2nd floor]

→ All these switches act as access layer which help to provides network connectivity to all room, lab, etc.

* It's directly connected to:-

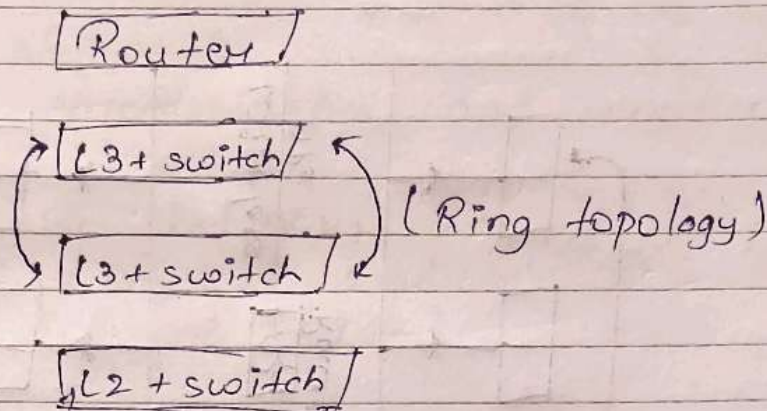
① Ethernet (RJ45)

② Wireless

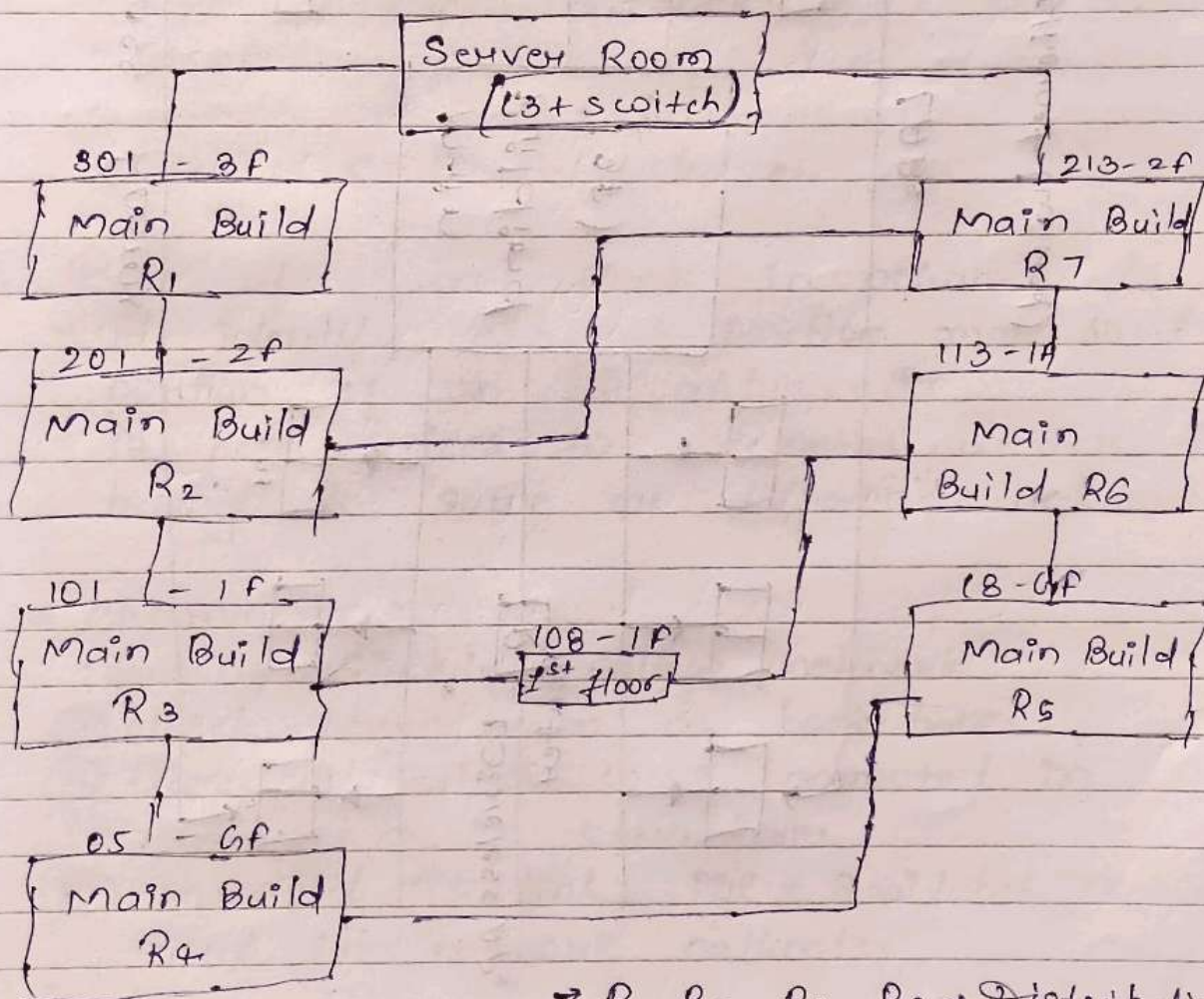
③ Lab system

④ Outdoor wireless.

* Server Room Architecture / Topology

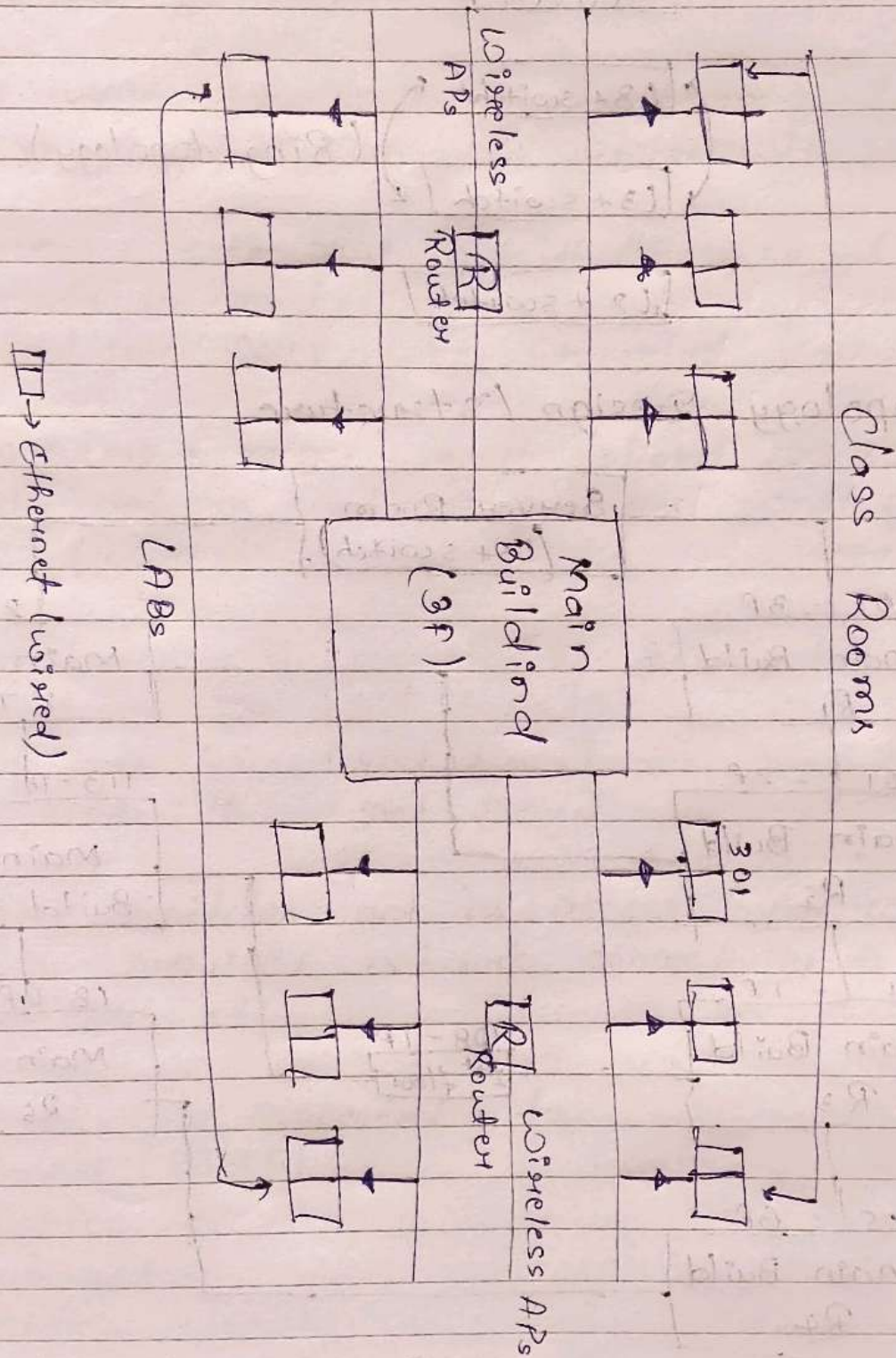


* Topology Design / Structure



- R, R7, R2, R6 → Distribution layer
- Server room → Core layer
- R1 - R7 → Access layer

* Access layer Switch (3rd floor - 301)



Q. Difference between wired and wireless speed?

→ Wired Network

- faster communication and connection
- Stable connection.
- Very-low latency.
- Direct / physical medium.
- Used - Ethernet (RJ45)

→ Wireless Network

- Slower connection / communication.
- Affected by distance.
- Higher latency.
- Shared medium radio.
- Used - Router to devices.

Q. Role of Subnet Mask in network.

- It identify network portion and host portion of an IP address. It
- It help devices to determine whether, they belong to same or different network.

* Observation:

→ During studying college network

① Used server room as back bone.

② Two L3 switch were connected in reduced cyclic structure.

③ managed TP-Link PoE + Switches were used in network cabinets.

* Advantage of Existing Network distribution:

- Scalable Architecture
- Centralized managed
- Support POE+ devices
- Easy expansion of floor & class room.
- Support both wired and wireless.

* Conclusion

The campus LAN follows a hybrid hierarchical topology consists of tree, Extended star Ring. The network used to managed by POE switches wired APs, Ethernet & centralized server infrastructure and efficient wired and wireless communication across the campus.

IP address, Subnet Mask and MAC address using command "ipconfig"

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Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\ashis> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 10:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::857e:8349:2826:8723%12
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    IPv4 Address. . . . . : 10.101.28.144
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.101.28.149

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
```

Wi-Fi details by using command "netsh wlan show interfaces"

```
Windows PowerShell
PS C:\Users\ashis> netsh wlan show interfaces

There is 1 interface on the system:

    Name                : Wi-Fi
    Description          : Realtek 8852BE Wireless LAN WiFi 6 PCI-E NIC
    GUID                 : 34821c38-2de0-4002-827b-910e0b351e3d
    Physical address     : 94:bb:43:82:86:f6
    Interface type       : Primary
    State                : connected
    SSID                 : krshriraunak
    AP BSSID             : 56:d3:9e:8c:f6:7f
    Band                 : 2.4 GHz
    Channel              : 11
    Connected Akm-cipher : [ akm = 00-0f-ac:08, cipher = 00-0f-ac:04 ]
    Network type         : Infrastructure
    Radio type           : 802.11n
    Authentication       : WPA3-Personal
    Cipher               : CCMP
    Connection mode      : Auto Connect
    Receive rate (Mbps)  : 72.2
    Transmit rate (Mbps) : 72.2
    Signal               : 100%
    Rssi                 : -20
    Profile               : krshriraunak
    QoS MSCS Configured  : 0
    QoS Map Configured   : 0
    QoS Map Allowed by Policy : 0

PS C:\Users\ashis> |
```


Ethernet details by using command "get-NetAdapter"

```
Windows PowerShell
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PS C:\Users\ashis> netsh interface show interface
```

Admin State	State	Type	Interface Name
Enabled	Disconnected	Dedicated	Ethernet
Enabled	Connected	Dedicated	Ethernet 10
Enabled	Disconnected	Dedicated	Wi-Fi
Enabled	Connected	Dedicated	Ethernet 11

```
PS C:\Users\ashis> get-NetAdapter
```

Name	InterfaceDescription	ifIndex	Status	MacAddress	LinkSpeed
Bluetooth Network Conn...	Bluetooth Device (Personal Area Netw...	26	Disconnected	94-BB-43-82-86-F7	3 Mbps
Ethernet 11	Remote NDIS based Internet Sharing...#5	58	Up	BE-8C-33-D6-B9-D0	426.0 Mbps
Ethernet 10	VirtualBox Host-Only Ethernet Adapter	12	Up	0A-00-27-00-00-0C	1 Gbps
Ethernet	Realtek PCIe GbE Family Controller	11	Disconnected	CC-28-AA-15-BA-D1	0 bps
Wi-Fi	Realtek 8852BE Wireless LAN WiFi 6 P...	10	Disconnected	94-BB-43-82-86-F6	72.2 Mbps

```
PS C:\Users\ashis> |
```